Quantium Virtual Internship – Task 2: Trial vs Control Store Evaluation

```
library(data.table)
library(ggplot2)
library(tidyr)
theme_set(theme_bw())
theme_update(plot.title = element_text(hjust = 0.5))
```

1. Load Data

```
filePath <- "PATH/TO/YOUR/DATA/"
data <- fread(paste0(filePath, "QVI_data.csv"))
```

2. Create Monthly Metrics

```
data[, YEARMONTH := year(DATE) * 100 + month(DATE)]
measureOverTime <- data[, .(
totSales = sum(TOT_SALES),
nCustomers = uniqueN(LYLTY_CARD_NBR),
nTxnPerCust = .N / uniqueN(LYLTY_CARD_NBR)
), by = .(STORE_NBR, YEARMONTH)][order(STORE_NBR, YEARMONTH)]
```

3. Filter to Pre-Trial Period

```
storesWithFullObs <- unique(
measureOverTime[, .N, STORE_NBR][N == 12, STORE_NBR]
)
preTrialMeasures <- measureOverTime[
YEARMONTH < 201902 & STORE_NBR %in% storesWithFullObs
]
```

4. Functions for Control Store Selection

4.1 Correlation Function

```
calculateCorrelation <- function(inputTable, metricCol, trialStore) {
controlStores <- unique(inputTable[, STORE_NBR])
output <- data.table()
for (store in controlStores) {
corr <- cor(
```

```
inputTable[STORE_NBR == trialStore, get(metricCol)],
inputTable[STORE_NBR == store, get(metricCol)]
)
output <- rbind(output, data.table(
Store1 = trialStore,
Store2 = store,
corr_measure = corr
))
}
return(output)
}
```

4.2 Magnitude Distance Function

```
calculateMagnitudeDistance <- function(inputTable, metricCol, trialStore) {
controlStores <- unique(inputTable[, STORE_NBR])
distTable <- data.table()
for (store in controlStores) {
dist <- abs(
inputTable[STORE_NBR == trialStore, get(metricCol)] -
inputTable[STORE_NBR == store, get(metricCol)]
)
distTable <- rbind(distTable, data.table(
Store1 = trialStore,
Store2 = store,
measure = mean(dist)
))
}
distTable[, mag_measure :=
1 - (measure - min(measure)) / (max(measure) - min(measure))
]
return(distTable[, .(Store1, Store2, mag_measure)])
}
```

5. Trial Store 77 – Control Selection

```
trial_store <- 77
```

```
corr_sales <- calculateCorrelation(preTrialMeasures, "totSales", trial_store) corr_cust <-
calculateCorrelation(preTrialMeasures, "nCustomers", trial_store)

mag_sales <- calculateMagnitudeDistance(preTrialMeasures, "totSales", trial_store) mag_cust <-
calculateMagnitudeDistance(preTrialMeasures, "nCustomers", trial_store)

score_sales <- merge(corr_sales, mag_sales, by = c("Store1", "Store2")) score_sales[, score :=
0.5 * corr_measure + 0.5 * mag_measure]

score_cust <- merge(corr_cust, mag_cust, by = c("Store1", "Store2")) score_cust[, score := 0.5 *
corr_measure + 0.5 * mag_measure]

final_scores <- merge( score_sales[, .(Store2, score_sales = score)], score_cust[, .(Store2,
score_cust = score)], by = "Store2" )

final_scores[, finalScore := 0.5 * score_sales + 0.5 * score_cust] control_store <-
final_scores[order(-finalScore)][Store2 != trial_store][1]

control_store
```

6. Trial Impact Assessment – Store 77

6.1 Scale Control Sales

```
scalingFactor <- preTrialMeasures[

STORE_NBR == trial_store, sum(totSales)

] / preTrialMeasures[

STORE_NBR == control_store, sum(totSales)

]

scaledControl <- measureOverTime[

STORE_NBR == control_store,

.(YEARMONTH, controlSales = totSales * scalingFactor)]

]

comparison <- merge(

scaledControl,

measureOverTime[STORE_NBR == trial_store, .(YEARMONTH, trialSales = totSales)],

by = "YEARMONTH"

)

comparison[, pctDiff := (trialSales - controlSales) / controlSales]
```

6.2 Statistical Test

```
stdDev <- sd(comparison[YEARMONTH < 201902, pctDiff])
comparison[, tValue := pctDiff / stdDev]
```

6.3 Visualization

```
comparison[, Date := as.Date(paste0(YEARMONTH, "01"), "%Y%m%d")]
ggplot(comparison, aes(Date)) +
geom_line(aes(y = trialSales, colour = "Trial")) +
geom_line(aes(y = controlSales, colour = "Control")) +
geom_rect(
aes(xmin = as.Date("2019-02-01"),
xmax = as.Date("2019-04-30"),
ymin = -Inf, ymax = Inf),
fill = "grey80", alpha = 0.3
) +
labs(
title = "Trial vs Control Sales – Store 77",
y = "Total Sales",
colour = "Store Type"
)
```